Generative UI and the Reconfiguration of Design Practice

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Current research often frames GenUI as a technical challenge or a means of producing bespoke interfaces. In this position paper, we propose understanding GenUI as a multifaceted socio-technical phenomenon that reconfigures participatory design, redistributes roles traditionally held by designers and developers, and breaches established norms of software production. We argue that GenUI can introduce new skill demands beyond prompt engineering and raise unresolved questions of responsibility, accountability, and inclusion that warrant further theoretical and empirical attention.

CCS Concepts: • **Computing methodologies** → *Artificial intelligence*; • **Human-centered computing** → **Human computer interaction (HCI)**; **Interactive systems and tools**.

Additional Key Words and Phrases: Generative UI, AI, Design

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1 Introduction

The idea of automatically generating user interfaces is not particularly new [7]. However, with the recent advances in AI technology, a new approach, dubbed Generative User Interfaces (GenUI), has emerged [6]. GenUI is commonly framed as both a novel way to create bespoke interfaces and applications [5] and a technical challenge [2]. In this position paper, we argue that such a framing is insufficient and GenUI should be treated as a multifaceted phenomenon that simultaneously operates as a design practice, a technical challenge, and an industry disruption.

2 Perspectives on GenUI

From a design perspective, GenUI can be seen as an *extreme form of participatory design* [1], where the user involvement is maximized, and the human designer is replaced by AI. Traditional participatory design involves users contributing feedback, ideas, or domain knowledge to a design process led by professional designers. With GenUI, users can become moment-to-moment designers, actively (re-) shaping the interface in real time. This shifts participation and user involvement from a staged, curated activity into a continuous practice that is not guarded or governed by protocols and time-limited activities.

At the same time, this "extreme participation" will not necessarily be frictionless or universally empowering. It requires new competencies, such as prompt articulation and conceptual modelling of tasks, intents, and activities.

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Professional designers, who are trained to guide and facilitate the design process and usually have the knowledge and understanding of design guidelines and best practices, are excluded from this process.

Another important role of a professional designer is that of a *reflective practitioner* [8], who actively studies the domain area, practices, conventions, and specific user settings during the participatory design process. Modern AI models, which would replace designers in the GenUI approach, are incapable of on-the-fly retraining, making it impossible for them to actually "learn", only accumulate data.

Seen technically, GenUI could introduce challenges that go well beyond model accuracy or interface generation quality [2]. Generating an interface is not equivalent to generating an image or a paragraph of text. Interfaces are operational artifacts: they must be coherent over time, support error recovery, maintain internal consistency, and align with both user intent and system constraints. GenUI systems must include reasoning about interaction flow, state, dependencies, and consequences.

Moreover, GenUI systems must align with existing software and are inherently part of a socio-technical ecosystem [4], legacy software solutions, and organizational constraints. The technical problem is therefore not simply to “generate a UI” but to “generate a situated, accountable, and maintainable interactive tool.” Therefore, AI models must replace not only designers but also software engineers, architects, and product owners who are familiar with the specifics of the software, data formats, communication and exchange protocols, and various requirements that are not necessarily obvious to end users. This places GenUI as a field of study at the intersection of machine learning, software architecture, HCI/CSCW, and systems engineering, rather than within any one of them.

GenUI can also be interpreted as a *breaching experiment* [3] within the software design and development industry. Like classic breaching experiments in sociology, GenUI exposes tacit norms by violating them. It challenges the assumed division of labor between designers, developers, and users and questions the idea that interfaces must be fixed artifacts finalized before use and versioned through controlled releases. GenUI raises questions like: Who is accountable for the generated interface? Who makes the design decisions? By breaching these norms, GenUI makes visible the social and organizational structures that have long been taken for granted in software production.

3 Conclusion

Taken together, these perspectives suggest that GenUI should not be evaluated solely on efficiency, usability, or novelty. It should be examined as a socio-technical shift that reconfigures participation, challenges professional norms, raises new technical problems, and gestures toward alternative futures of interaction. This shift raises a further question: *What additional skills must users acquire beyond prompt engineering to effectively participate in GenUI-based design?* As the roles traditionally occupied by designers, software developers, and system architects are partially redistributed between AI systems and end users, it is reasonable to expect users to assume some of the competencies associated with professional design and development practice. These may include the ability to reason about trade-offs, anticipate downstream consequences of design decisions, recognize breakdowns in interactions, plan the implementation of features, and identify software errors and ways to fix them. Importantly, users are not expected to become experts in these fields. Instead, GenUI asks users to apply parts of design and engineering judgment only when needed, guided by their immediate goals and the specific situation they are in. This selective transfer of competence further complicates questions of responsibility, expertise, and inclusion, as different users will be unevenly positioned to acquire and exercise these skills.

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